

National University of Singapore
Semester Examination for the Degree of B.Sc.

(Semester I)

CZ3272 – Monte Carlo and Molecular Dynamics

Nov 2004 – Time Allowed: 2 hours

Instructions to Candidates:

1. This is a closed book examination.
2. This examination paper contains FIVE (5) questions and comprises THREE (3) printed pages.
3. All questions carry equal marks.
4. Answer ALL the questions.

1. (Ising Model & Probability) Consider a two-dimensional Ising model of $N=L \times L$ with nearest neighbor interaction. We assume that all the spins are in the up states, $\sigma_i = +1$ for all site i . If a standard Metropolis step is performed at temperature T with a site choosing at random,
- (a) What is the probability that a particular site, say site 1, is flipped?
 - (b) What is the probability that any site (we don't care which one) is flipped?
 - (c) What is the probability that the system does not change?

[20 marks]

(a) x/N , (b) x , (c) $1 - x$, where $x = \exp[-8J/(k_B T)]$.

2. (Random Variable) A continuous random variable $0 < x < \infty$ is distributed according to the probability density $P(x) = x e^{-x}$. Give a method to generate random variable x that follows this distribution from a uniformly distributed random number $0 < \xi < 1$.
- [20 marks]

Use inverse method, $F(x) = \int_0^x y e^{-y} dy = 1 - (1+x)e^{-x}$. So $\xi = (1+x)e^{-x}$, x is obtained by solving the equation. However, there is no analytical expression for the solution, so the method is not practically. One can use Metropolis algorithm as well.

3. (Monte Carlo Integration) Consider a simple Monte Carlo evaluation of the integral

$$I = \int_0^{\infty} (\sin x) x e^{-x} dx.$$

Answer the following:

- (d) What random variable should one use for an estimator of I ? [That is, the probability distribution of the random variable x]
- (e) Give the expression for an estimator of I [This will be an arithmetic average of some quantity].
- (f) Give an estimator for the Monte Carlo error [standard deviation] σ .

[20 marks]

(d) Use that proposed in problem 2, with probability distribution $P(x) = x \exp(-x)$. So

(e) $I = \frac{1}{N} \sum_{i=1}^N \sin(x_i)$ and

(f) $\text{var}(x) = \frac{1}{N} \sum_{i=1}^N \sin^2(x_i) - I^2$, $\sigma = \sqrt{\frac{\text{var}(x)}{N-1}}$.

4. (Markov Chain) A transition matrix W in a Markov chain is given by

$$\begin{bmatrix} 1/2 & 0 & 1/2 \\ 1/3 & 1/3 & 1/3 \\ 0 & 1 & 0 \end{bmatrix}.$$

The matrix element W_{ij} corresponds transition probability from state i to j , $i=1,2,3$, $j=1,2,3$. Answer the following questions:

- Is this Markov chain irreducible?
- Are there any absorbing states?
- If the system is initially in state 1, what is the probability that the system will be in state 3 after two steps?
- What is its invariant (or equilibrium) probability distribution?

[20 marks]

(a) Compute $W^2 = \begin{bmatrix} \frac{1}{4} & \frac{1}{2} & \frac{1}{4} \\ \frac{5}{18} & \frac{4}{18} & \frac{5}{18} \\ \frac{1}{3} & \frac{1}{3} & \frac{1}{3} \end{bmatrix}$, which is nonzero for all i and j , so it is irreducible.

(b) No absorbing states for irreducible Markov chain.

(c) This is 3rd component of $(1, 0, 0) W^2$, which is $1/4$.

(d) Solve the linear equation $P=PW$, with normalization $P_1+P_2+P_3=1$, we get $P=(2/7, 3/7, 2/7)$.

5. (Molecular Dynamics) (a) Show that the symplectic algorithm A is equivalent to Verlet algorithm. The symplectic algorithm A is:

$$p(t+h) = p(t) + hF(q(t))$$

$$q(t+h) = q(t) + \frac{h}{m} p(t+h)$$

- (b) Show that the above algorithm viewed as a transformation from $[p(t), q(t)]$ to $[p(t+h), q(t+h)]$ preserves the phase space volume.

[20 marks]

(a) Write the second equation twice, once for t , nice for $t-h$, we have

$$q(t+h) = q(t) + (h/m)p(t+h) \quad (a)$$

$$q(t) = q(t-h) + (h/m)p(t) \quad (b)$$

Subtract, (a)-(b), and using the first equation in algorithm A, we get

$$q(t+h) - 2q(t) + q(t-h) = (h/m)(p(t+h) - p(t)) = (h^2/m)F(q(t))$$

which is the Verlet algorithm.

(b) Let's call the values at $t+h$, the hated version, we got

$$\frac{\partial \hat{p}}{\partial p} = 1, \frac{\partial \hat{p}}{\partial q} = hF'(q), \frac{\partial \hat{q}}{\partial p} = \frac{h}{m}, \frac{\partial \hat{q}}{\partial q} = 1 + \frac{h^2}{m} F'(q),$$

$$\begin{vmatrix} 1 & hF' \\ h/m & 1 + h^2 F' / m \end{vmatrix} = 1$$

-- End --